

MICHAEL LO RUSSO

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SUMMARY

Engineering student at UNSW specialising in Mechatronics, with practical experience spanning embedded systems, mechanical design, software development, and construction. Built a residential house, autonomous robots, and 3D-printed thermal hardware. Available for full-time work while completing my degree.

Visit my website portfolio for more details on my projects: [mlorusso.com](#)

EDUCATION

Bachelor of Engineering (Honours) (Robotics and Mechatronics)

Jan 2022 – Aug 2026

UNSW | WAM: 75 (Distinction)

RESEARCH EXPERIENCE

Honours Thesis — PVT & Heat Pump Assessment Tool | UNSW / Coolsheet

Sep 2025 – Present

- Developing a web-based modelling tool (Python back-end, HTML/CSS front-end) that simulates energy performance and cost-saving potential of Coolsheet's photovoltaic-thermal (PVT) panels and heat pump systems for commercial and industrial applications.
- Delivered a working v1 calculator with modular architecture, integrating Australian climate data via weather APIs to generate location-aware performance estimates across multiple building types.
- Cross-validating simulation outputs against real-world installation data from Coolsheet's commercial PVT deployments.

ENGINEERING PROJECTS

Rubik's Cube Solving Robot

Mar 2026 – May 2026

- Engineered a 5-axis robotic manipulator (NEMA 17 steppers, TMC2209, ESP32) that accepts a scrambled cube state via a custom web interface and computes a solution in under 1 second using the Kociemba two-phase algorithm.
- Integrated full-stack system spanning Fusion 360 mechanical design, C++ firmware with stepper motor control, a Python solving server, and an HTML/JavaScript UI for cube face input.

Micromouse Maze Navigation Robot | UNSW

May 2024 – Aug 2024

- Designed and programmed an autonomous maze-solving robot using LiDAR, IMU, and wheel encoders with PID control; navigated a 3.6 x 2.4 m arena with walls and obstacles in under 90 seconds.
- Implemented BFS path planning with OpenCV occupancy grid mapping on an Arduino Nano, processing sensor data in real time for autonomous navigation.

Residential Construction

Jun 2024 – Apr 2025

- Constructed a residential house alongside my father, executing framing, roofing, plumbing, and electrical work while ensuring all stages adhered to building compliance and regulatory standards.

Custom GPU Cooling Funnels

Jan 2025

- Designed and 3D-printed airflow funnels in Fusion 360 across 3 design iterations; reduced GPU temperatures by 7°C under full load, inspired by automotive ducting principles.

UR5e Robotic System | UNSW

May 2025 – Aug 2025

- Programmed a UR5e industrial robot for autonomous math problem solving, pick-and-place, and orientation adjustment. Developed trajectory planning to produce smooth pen lifts and legible handwriting output.

Unmanned Ground Vehicle Control System | UNSW

Sep 2024 – Dec 2024

- Built a multi-threaded C++ control system integrating LiDAR, GNSS, Xbox controller, and collision sensors for autonomous and teleoperated driving with real-time crash avoidance and MATLAB visualisation.

WORK EXPERIENCE

Experience Consultant | Samsung Electronics (Parramatta)

Jan 2023 – Present

- Diagnosed and resolved hardware/software issues across Samsung mobile, tablet, and smart device lines, consistently maintaining a net promoter score above 85%.
- Generated \$950k+ in sales revenue while communicating complex technical information to both technical and non-technical customers across in-store and phone channels.
- Trained new staff on product knowledge and store procedures. Managed daily store opening and closing operations and assigned sales budgets to team members.

TECHNICAL SKILLS

Languages: C, C++, Python, Java, VHDL, HTML/CSS

Systems & Control: Sensor fusion (LiDAR, GNSS, IMU), PID control, real-time path planning, OpenCV, Arduino, MATLAB/Simulink

CAD & Manufacturing: SolidWorks, Fusion 360, Bambu Studio, 3D printing (FDM/PLA)

Web & Tools: Next.js, TypeScript, Tailwind CSS, Git, VS Code, LaTeX

AI & Automation: Claude Code, Codex, Cursor, Vercel, Antigravity, n8n